Bitcoin: A Peer-to-Peer Electronic Cash System. Response. Xiyao Long

The paper proposes a solution to allow payee check whether an owner of an electronic coin has double spent the coin. The proposed solution is to publish the hash of a block(transaction) to a broad network of nodes, and each of these nodes would attempt to work on solving a “proof of work” problem basically by using its own computational power. Whoever wins can generate the next block and at the same time earn some new coin. In this process, all the current block generated are hashed with the hash of previous blocks, therefore all the blocks are chained together. This structure means that any block in the middle can’t be redo without redoing all the blocks after it.

The node solving the problem will broadcast the added block to all the nodes in the network and the accepting nodes will decide whether to accept it or not, based on verifying no transactions in it involves double-spending. A node will also consider the longest blocks to be the correct one and continue to work to extend it rather than any other version. I think this design somehow is analogous to the Raft system/protocol, the members will consider a log with the highest term number and log size to be the most recent and correct. In Raft, the correctness a new entry is verified by the matching of term number and log index of its previous entry. Here in this system, the new block is also having its identity reinforced by its previous blocks. What’s different is that there’s not a leader and not even an elected leader such as in Raft. Here, a node gains the power by solving a problem “proof of work”. I think this may be due to that such block generation is a relative infrequent compared to the processing of messages as entries in the Raft system. Therefore we don’t need the election system to circumvent the disputes of complete decentralized system.

The author also talks about how the system can sustain the generation of new coins and offer incentive for people to stay honest in the system without any centralized agency: People can own new coins if they spent computational power to work on generating a new block; And the output value of transaction can be different from its input value so people are motivated in owning such coins.

I think the author can more explicitly talks about what happens if a receiving node finds out that a transaction in a received block involves double spending. This will give a better explanation about how it verifies double-spending.